

①

Orthogonal matching pursuit (OMP)

algorithm pseudocode

Input: $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$

Output: sparse representation $x \in \mathbb{R}^n$

1. $x_0 := 0$, $r_0 := b$, $S_0 := \emptyset$, $k := 0$
2. While (# of iterations ≤ 100) do

$$g_k := A^T r_k$$

$$j_{k+1} := \text{supp}(H_1(g_k))$$

$$S_{k+1} := S_k \cup \{j_{k+1}\}$$

$$(x_{k+1})_{S_{k+1}} := A_{S_{k+1}}^+ b, \quad (x_{k+1})_{S_{k+1}^c} := 0$$

$$r_{k+1} := b - A x_{k+1}$$

$$k := k + 1$$

Notations:

[1] H_S - the hard thresholding operator

$$H_S: x \mapsto H_S(x) \in \mathbb{R}^n$$

The vector $H_S(x)$ sets all entries of x to zero except for the
 (5) largest (in abs. value) entries

[2] A^+ is the pseudo-inverse of A . It is defined by means of the singular value decomposition (SVD)

(3)

Q SVD:

$$A_{m \times n}, \text{rank}(A) = r.$$

$$A = U \Sigma V^T$$

where ~~U, V~~ $U_{m \times m}$ and

$V_{n \times n}$ are orthogonal matrices

and $\Sigma_{m \times n}$ is a diagonal matrix with non-zero diagonal entries

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$$

Let $\Sigma^+_{n \times m}$ be the diagonal matrix

$$\Sigma^+ = \begin{pmatrix} \frac{1}{\sigma_1} & & & & & \\ & \ddots & & & & \\ & & \frac{1}{\sigma_r} & & & \\ & & & & 0 & \\ 0 & & & & & \\ & & & & & 0 \end{pmatrix}$$

④

Pseudo inverse is
defined by

$$A^+ := (V \Sigma^+ U^T)_{n \times m}$$